

HARSH DABHI

Senior Software Engineer | AI Systems & Forward-Deployed Engineering

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PROFESSIONAL SUMMARY

Senior Software Engineer with 5+ years shipping production AI systems, LLM applications, and the backend platforms that run them. Work spans RAG pipelines, multi-agent orchestration, and high-throughput data infrastructure—deployed for enterprise teams at Blue Yonder, Dentsu, and Select Equity Group across SRE, workflow automation, and financial data domains.

Currently a Forward-Deployed AI Engineer embedded directly in customer teams, converting ambiguous requirements into deployed production systems. Builds RAG pipelines, multi-agent workflows, and LLM-powered automation across OpenAI, Claude, and Gemini, backed by scalable services in Python, FastAPI, and Django.

Works across the full AI product lifecycle, from model orchestration and evaluation to APIs, Kubernetes, and cloud infrastructure, with a track record of taking systems from customer requirements to production deployment.

WORK EXPERIENCE

Calidad Infotech | SDE-2 (Software Development Engineer II)

Sep 2023 – Present

- Embedded within customer teams at Blue Yonder and Dentsu as a Forward-Deployed AI Engineer, shipping LLM-powered systems that automated incident response and campaign operations, removing recurring manual work from live engineering and ops workflows.
- Architected ozar.ai, an LLM-powered test-case generation SaaS on AWS (Lambda, RDS, API Gateway, Terraform), incorporating a RAG pipeline with RLHF scoring loop, a no-code automation engine, and multi-agent orchestration that turns a Jira ticket into ready-to-run test cases without an engineer writing them by hand.
- Built testautomationx.com end to end, an isolated test execution engine backed by AWS SQS and Celery that scales to 1000+ concurrent jobs executing in parallel, deployed on Kubernetes with Docker, GitHub Actions CI/CD, Selenium, and PyTest.
- Delivered two AI systems for client Data Processing Forms Pvt Ltd — a multi-LLM question generation framework (OpenAI GPT, Claude API, Gemini via LangChain) and a ResNet-based multilingual OCR classifier achieving ~80% accuracy across handwriting scripts.
- Led prompt engineering and RAG pipeline design across several client and internal projects, grounding OpenAI, Claude, and Gemini outputs in retrieved source data and adding output validation so generated responses stayed traceable to evidence rather than free-form model guesses.
- Migrated the internal product stack from AWS to a self-managed Hostinger Kubernetes cluster built from scratch, reducing deployment costs by 60–70% while adding centralised logging and Grafana monitoring.
- Designed and shipped production marketing sites for testautomationx.com, ozar.ai, and calidadinfotech.com using AI coding assistants, delivering all three sites without a dedicated frontend engineer.
- Recognised with the Code Innovator Award in both 2024 and 2025 for technical work within Calidad Infotech.

XDuce Pvt. Ltd. | Python Developer → Software Engineer

Jun 2022 – Sep 2023

- Built a real-time financial data ETL platform for Select Equity Group, ingesting 10,000+ tickers per second via WebSockets into Snowflake and BigQuery, normalising data across 50 tables from 4–5 source pipelines with cross-source data quality checks to eliminate ambiguity.

- Developed ML anomaly detection and predictive analytics models with scikit-learn, deployed behind REST APIs on Docker with Azure DevOps CI/CD, flagging outliers in incoming financial data before they reached downstream reports.
- Designed complex SQL reconciliation queries across financial datasets and containerised all services on GCP with Docker, backed by comprehensive PyTest suites that caught data integrity issues before production.
- Ran delivery of multiple concurrent data engineering and projects in Agile sprints, coordinating across cross-functional teams to ship each on its committed timeline.
- Received the Rising Star Award (2022), given in first year for delivery of the Select Equity Group ETL platform and onboarding of new pipeline technologies.

Sep 2020 – Jun 2022

Riddhi Corporate Services Limited | *Junior Software Engineer*

- Built business data automation and reporting solutions with .NET, Python, SQL Server, and VBA that replaced recurring manual report-building, converting multi-hour weekly analyst tasks into scheduled jobs that ran without intervention.
- Designed data visualisation dashboards and ETL scripts that consolidated feeds from multiple disconnected business systems into a single reporting view, replacing manual cross-system collation for senior management reviews.
- Translated business requirements from cross-functional stakeholders into production-ready software solutions, delivering full design, development, and testing documentation for each engagement.

PROJECTS

AI-Powered SRE & Root Cause Analysis Platform — *Client: Blue Yonder | Calidad Infotech*

Tech: *Python, Azure OpenAI, LangGraph, Langfuse, RAG, Zabbix, AppDynamics, FastAPI, Docker, Azure*

- Engineered an AI-driven RCA system that reduced incident resolution time from several hours to under 20 minutes, an 80%+ improvement that directly cut on-call burden for the SRE team.
- Ingested observability data from Zabbix, AppDynamics, and internal DBs into a RAG pipeline, enabling the system to surface actual root causes rather than surface-level symptoms.
- Orchestrated a LangGraph multi-agent flow on Azure OpenAI to execute the correct SOP per incident, isolate the fault, and auto-generate a structured RCA — traced and evaluated end to end with Langfuse.
- Deployed the full stack on Azure with Docker and wired FastAPI endpoints directly into the client's SRE dashboards and on-call tooling, making RCA output immediately actionable.

AI Workflow Automation & Forward-Deployed Engineering — *Client: Dentsu | Calidad Infotech*

Tech: *Python, OpenAI GPT, Claude API, LangChain, FastAPI, REST APIs, Docker, Workflow Orchestration*

- Embedded within Dentsu's teams to map automation opportunities across business units and ship LLM-powered tools that removed manual steps from live operational and campaign workflows.
- Designed end-to-end AI workflow automation pipelines spanning content generation, data enrichment, and approval routing, collapsing multi-step manual processes into single triggered runs and removing the hand-offs where errors crept in.
- Engineered reusable prompt frameworks that enabled Dentsu's non-ML teams to adopt AI tooling independently, reducing dependency on engineering for routine AI tasks.
- Connected the pipelines to Dentsu's existing third-party platforms (CRM, ad, and analytics APIs), so automated outputs flowed straight into the tools campaign teams already used instead of routing through a manual export-and-upload step.

ozar.ai — AI Test Case Generation SaaS Platform — *Calidad Infotech (Internal Product)*

Tech: *Django REST Framework, LangGraph, RAG, RLHF, Multi-Agent Orchestration, AWS, Celery, Docker*

- Architected an LLM-powered SaaS that generates context-aware test cases directly from PMS/TMS sources such as Jira, cutting manual test-writing effort for engineering teams.
- Implemented a RAG pipeline with an AI scorer and an RLHF feedback loop, using reranking to tune the precision/recall trade-off and few-shot anchoring so test-case output held the same structure and coverage from one generation run to the next.

- Built a no-code automation engine that lets users describe a test flow in plain English, automates the scenario end to end, and exports a ready-to-run structured automation script — removing the need to hand-code test cases.
- Built and shipped frontend features for the product, contributing across the stack despite not having a dedicated frontend engineer.
- Deployed the backend serverless on AWS (Lambda, RDS, API Gateway via Terraform) with Celery for async LLM processing and Razorpay for subscription billing, keeping infrastructure costs low while supporting concurrent users.

testautomationx.com — Test Case Management Platform — Calidad Infotech (Internal Product)

Tech: Python, Django REST Framework, AWS SQS, Celery, Kubernetes, Docker, Selenium, pytest, GitHub Actions

- Designed and delivered testautomationx.com end to end — a full test-case management platform with APIs for creation, categorisation, execution triggering, and reporting across the entire test lifecycle.
- Engineered an isolated execution engine on AWS SQS and Celery that runs automation jobs asynchronously in true parallel, scaling to 1000+ concurrent jobs with no cross-run interference.
- Deployed on Kubernetes with Docker-containerised services for horizontal scaling and role-based access control, supporting secure multi-team usage at scale.
- Automated CI/CD pipelines via GitHub Actions and integrated Allure Reports for execution analytics, giving teams full visibility into test results and failure trends.

GenAI Multi-LLM Question Generation Framework — Client: Data Processing Forms Pvt Ltd | Calidad Infotech

Tech: Python, FastAPI, LangChain, OpenAI GPT, Claude API, Google Gemini API, Hugging Face, RAG

- Architected a multi-LLM routing framework that dispatches prompts across OpenAI GPT, Claude API, and Gemini using a custom model-ranking algorithm, selecting the best model per query type.
- Built modular prompt templates for MCQ, descriptive, and scenario-based question types with LangChain routing and schema validation, so every format returned well-formed, answer-keyed output ready to consume without manual cleanup.
- Developed the FastAPI backend and integrated Hugging Face models (including high-context models like NVIDIA's NIM) for semantic similarity scoring, automatically grading generated answers against reference keys instead of manual review.

ResNet-Based Multilingual OCR & Handwriting Classification — Client: Data Processing Forms Pvt Ltd | Calidad Infotech

Tech: Python, TensorFlow, Keras, ResNet, Tesseract OCR, OpenCV, scikit-learn, NumPy, Pandas, Matplotlib

- Trained a ResNet-based deep learning model for multilingual handwriting classification, achieving ~80% accuracy across multiple scripts and an F1 score in line with industry benchmarks for multi-class OCR tasks.
- Built a preprocessing pipeline covering normalisation, augmentation, and noise reduction, with Tesseract OCR as the baseline layer, improving raw model input quality and boosting downstream classification accuracy.
- Evaluated model performance using F1 score, confusion matrix, and per-class accuracy metrics, iterating on architecture to close gaps and improve generalisation to real-world handwriting samples.

Intelligent Eye-Motion Anti-Cheating System — Calidad Infotech (R&D)

Tech: Python, OpenCV, TensorFlow, Flask, WebRTC, WebSockets

- Built a real-time gaze and motion detection pipeline with OpenCV and TensorFlow that monitors live online assessments and flags suspicious behaviour as it happens.
- Integrated ML-based anomaly detection for real-time inference and built an event reporting module that logs every suspicious event with a timestamp, giving exam administrators a clear audit trail.
- Architected a low-latency live video streaming layer using Flask and WebRTC, enabling real-time inference without buffering delays that would break the proctoring experience.

Real-Time Financial ETL & Analytics Platform — Client: Select Equity Group | XDuce

Tech: Python, Snowflake, Google BigQuery, WebSockets, Jenkins, JAMS, GCP, Docker, Pandas, NumPy, pytest, SQL, REST APIs

- Designed scalable ETL pipelines ingesting 10,000+ financial tickers per second via WebSockets for Select Equity Group, normalising data across 50 tables from 4–5 source systems.
- Integrated Snowflake and BigQuery for warehousing across 50 normalised tables and built REST APIs exposing processed data to analytics consumers, with cross-source data quality checks that eliminated data ambiguity.

- Automated deployments with Jenkins and JAMS, containerised all services on GCP with Docker, and enforced data integrity through PyTest unit tests that caught pipeline regressions before they reached production.

Cloud Migration & DevOps Infrastructure — *Calidad Infotech (Internal)*

Tech: *Kubernetes, Docker, Hostinger, AWS, Grafana, Centralised Logging, CI/CD, HTML/CSS/JS*

- Migrated the entire internal product stack from AWS to a self-managed Hostinger server, standing up a Kubernetes cluster from scratch and cutting infrastructure deployment costs by 60–70%.
- Configured centralised logging and Grafana dashboards across all services, giving the team a single observability layer and reducing time to detect and investigate incidents.
- Designed and shipped production marketing sites for internal products without a dedicated frontend engineer.

TECHNICAL SKILLS

AI / LLM	LangChain · LangGraph · RAG · RLHF · Multi-Agent Orchestration · LLM Systems · Prompt Design · Evals · OpenAI · Claude API · Gemini · Hugging Face · Langfuse
Backend	Python · Django REST Framework · FastAPI · Celery / SQS · REST APIs · WebSockets · SSO / OAuth
Infrastructure	AWS (Lambda · RDS · SQS · API Gateway) · Azure / Azure OpenAI · GCP · Kubernetes · Docker · Terraform · GitHub Actions · Grafana · Jenkins / JAMS
ML / Vision	TensorFlow · Keras · ResNet · scikit-learn · OpenCV · Tesseract OCR · Pandas · NumPy · Matplotlib
Data	PostgreSQL · SQL · NoSQL · Snowflake · BigQuery · ETL Pipelines · Zabbix · AppDynamics
Frontend	React · Next.js · TypeScript · JavaScript · PyTest · Selenium · Allure · Git / GitHub · Agile

EDUCATION

Gujarat Technological University

2020

Bachelor of Engineering (B.E.) – Information Technology
Government Engineering College, Bhavnagar | CGPA: 7.97 / 10

CERTIFICATIONS

- Machine Learning from Basic to Advanced (2023)
- AWS for Solution Architecture (2023)
- DevOps Fundamentals (2022)
- Crash Course on Python — Google (2022)

AWARDS & ACHIEVEMENTS

- Code Innovator Award — Calidad Infotech (2025): Awarded for building the company's LLM and RAG product capability and deploying it into client engagements at Blue Yonder and Dentsu.
- Innovator Award — Calidad Infotech (2024): Awarded for architecting the ozar.ai and testautomationx.com internal products from the ground up.
- Rising Star Award — XDuce Corporation (2022): Awarded in first year for delivering the Select Equity Group real-time financial ETL platform across cross-functional teams.